

CHAPTER 11

Sensors

11.1. Space story

On the BSat launch in Japan, we were doing spin-axis attitude determination. Every time the spinning spacecraft saw the Sun it would measure the angle. We got files of angles. During transfer orbit, the spacecraft spin-axis was pointed in different directions. At one point we noticed a jump in Sun angle. The way the sensor worked was it had Sun detectors in a row and as the Sun moved along the row it would illuminate a different sensor giving an angle. Our customer ACS engineer realized that one of the detectors must be damaged so that its angle was never read. This was easily accommodated in the spin-axis attitude determination process, which was ground based.

11.2. Introduction

The performance of a spacecraft-control system is limited by the performance of its sensors and actuators. This chapter discusses the types of sensors that are used in spacecraft-control systems. Many of the sensors are used for determining the attitude or attitude rates of the spacecraft. Others are used for determining the relative orientation or position of components on a spacecraft.

11.3. Types of sensors

Table 11.1 lists the classes of sensors used in spacecraft. Within each class, there may be many types of sensors.

Table 11.1 Classes of sensors.

Sensor	Description	Application
Accelerometer	Measures acceleration on an element.	Inertial acceleration measurements, vibration sensing, nutation measurements.
Angle encoder	Measures either the absolute or relative angle of a shaft.	Angle measurements or angular-rate measurements.
Earth or planetary sensor	Measures the location of the sensor boresight relative to the nadir vector. Does not require the spacecraft to be rotating.	Orientation with respect to a planet.

continued on next page

Table 11.1 (*continued*)

Sensor	Description	Application
Gyro	Any device that measures angular rates.	Rotation angle and rate measurements.
Hall effect	Measures the change in a magnetic field.	Used to determine rotor position in a motor.
Horizon sensor	Detects the Earth's horizon. Requires spacecraft rotation.	Attitude determination for spinning spacecraft. Planetary width for navigation.
Magnetometer	Measures the vector magnitude of a magnetic field.	Attitude determination. Magnetic control.
Potentiometer	A variable-resistance device in which the resistance varies as it is rotated.	Measurement of angles between rotating spacecraft components.
Rangefinder	Determines the distance between two objects that are not connected.	Docking, orbit maneuvers.
Star mapper	Measures the times of the crossing of a star across a detector.	Inertial attitude determination.
Star tracker	Measures the position of a star image in the focal plane of the sensor.	Inertial attitude determination.
Sun sensor	Either a single-axis or two-axis measurement of the angle of the Sun with respect to the boresight of the sensor.	Attitude determination, in particular yaw-attitude determination. Also used for Sun-safe modes.

11.4. Planetary optical sensors

This section covers optical sensors that image a planet or other distributed body.

11.4.1 Horizon sensors

A horizon sensor uses the spacecraft to scan the horizon of a celestial body. The measurement geometry for the Earth is shown on the left in Fig. 11.1. The horizon sensor has a very narrow field of view and responds to the scanning of the Earth as shown below. The ideal Earth pulse, which is the time history of the sensor that shows the heat from the Earth, is shown on the right of Fig. 11.1.

Several problems can arise. If a bright object is near the main object, the scan will be extended, causing the sensor to think it is closer to the equator of the body than it is. A second problem can occur if there is a dim spot on the scanned body. The sensor may then decide the edge of the body has been detected. The shortened chord makes it think the scan is nearer the pole.

Generally, the main problem is detecting the edge. The Earth's CO₂ layer is about 40 km above the surface of the Earth. The layer is not constant and varies with the season of the part of the Earth being scanned. For “precise” attitude determination using this

kind of sensor, it is necessary to maintain correction factors for seasonal variations in the edge. The CO₂ layer is used because, despite these issues, it is more reliable than using the visible bands.

Horizon sensors are used for attitude determination on spinning or dual-spin spacecraft. Many spacecraft spin in transfer orbit and a horizon sensor, in conjunction with a single-axis Sun sensor, can provide complete spin-axis orientation information. Another example is the GGS Polar spacecraft that used a horizon sensor on the spun point for pitch-orientation measurements during its mission.

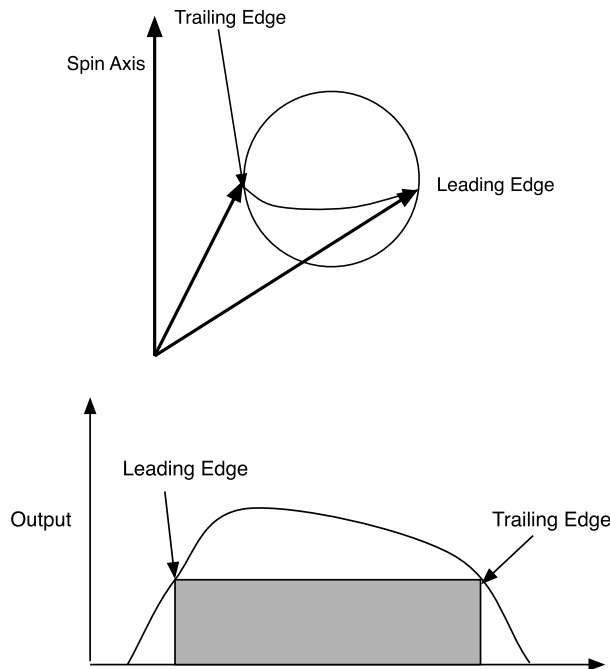


Figure 11.1 Horizon sensor measurement geometry and pulse.

11.4.2 Earth and planetary sensors

Earth and planetary sensors are sensors that detect the edge of the planet without requiring the spacecraft to spin. These also use an imager that sees the CO₂ band. There are three types: torsion-bar scanning, array, and conical scanning. The first uses an oscillating mirror to scan the Earth. The latter is similar to a horizon sensor. These sensors use the chordwidths to determine the roll attitude and the time from scan center to scan edge to determine pitch.

Scanning sensors measure roll by measuring the planetary chord at a specific cant angle and differencing the Earth-chord measurements the measurements to get an ap-

proximation to roll. The measured roll angle for a scanning sensor is

$$\theta_{meas} = 2 \left[\cos^{-1} \left(\frac{\cos \rho - \sin \delta \sin \theta}{\cos \delta \cos \theta} \right) - \cos^{-1} \left(\frac{\cos \rho + \sin \delta \sin \theta}{\cos \delta \cos \theta} \right) \right] \quad (11.1)$$

where δ is the angle of the scan as measured from the plane normal to the sensor scan axis positive about the roll axis, ρ is the angular radius of the Earth, and θ is roll. This relationship can be proven by assuming that

$$\gamma + \theta = \cos^{-1} \left(\frac{\cos \rho - \sin \delta \sin \theta}{\cos \delta \cos \theta} \right) \quad (11.2)$$

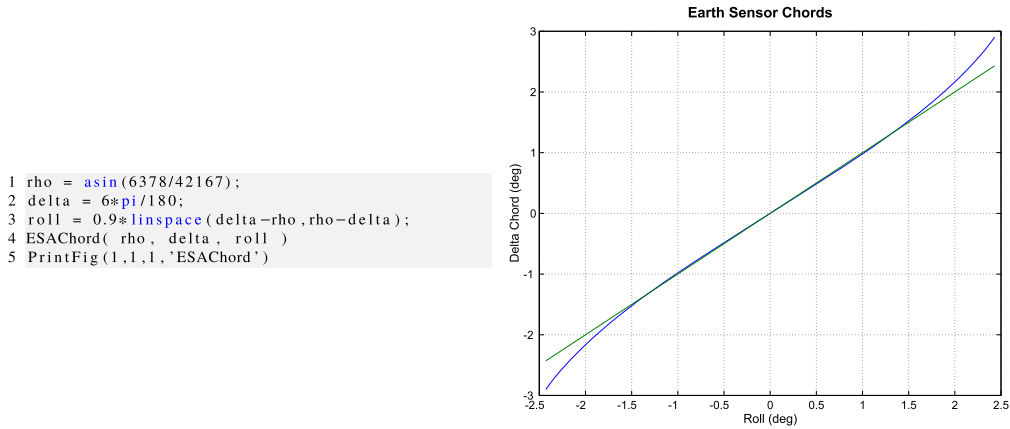
Taking the cosine of both sides and assuming that θ is small

$$\cos \gamma - \theta \sin \gamma = \frac{\cos \rho}{\cos \delta} - \theta \frac{\sin \delta}{\cos \delta} \quad (11.3)$$

This is true if $\delta \approx \rho$ so that

$$\theta_{meas} \approx \theta \quad (11.4)$$

This is shown in Example 11.1. The approximation is excellent except at the very ends of the range.



Example 11.1: Earth-sensor chord.

A major advantage of these sensors is that they directly give roll and pitch outputs. The range of this type of sensor can be enhanced by defining a standard chord, a moving average of the North and South chords. When both chords are on the Earth

$$\theta = \frac{1}{2} (\Omega_s - \Omega_n) \quad (11.5)$$

Then, when the South chord goes off the Earth roll can be approximated by

$$\theta = \Omega_a - \Omega_n \quad (11.6)$$

or when the North chord goes off the Earth

$$\theta = \Omega_s - \Omega_a \quad (11.7)$$

The sensor, with this simple logic, provides the correct sign while at least one chord is on the Earth, as shown in Example 11.2. There is also a roll-reversal region that can prove dangerous if a controller tries to use it to correct roll not knowing that the sign is reversed.

The static Earth sensor has sets of thermopiles arranged in a circle about the bore-sight. Typically each set has three thermopiles. One looks at the Earth all of the time, the second straddles the Earth, and the third looks at “cold” space. The former and latter are used to calibrate the straddling sensor. The calibrated temperatures of the straddling sensors are used to compute roll and pitch.

The second Earth sensor detects the edges with a fixed set of sensors in a circular array. This type can use solid sensing elements that ride on the edge of the Earth or use an array to determine the edge. The conical scanning Earth sensor has a motor that rotates a mirror or prism that sweeps the field of view of the sensing element across the Earth, see Fig. 11.2.

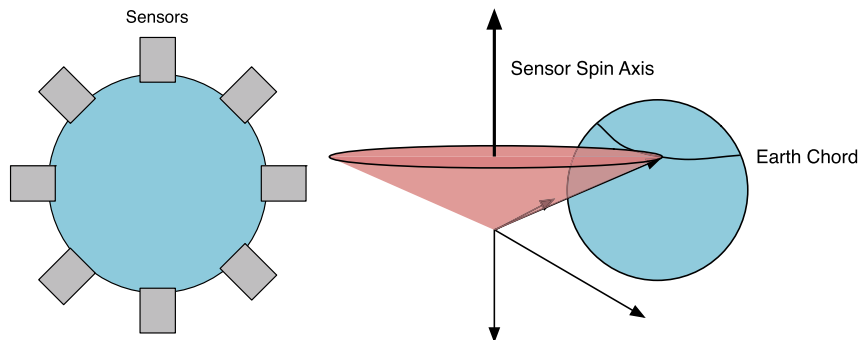


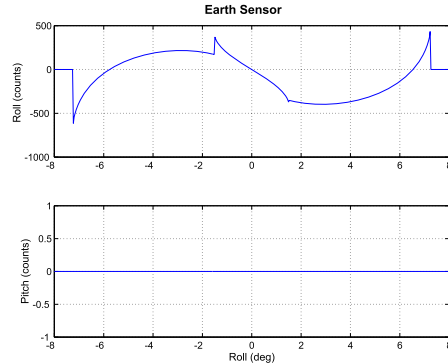
Figure 11.2 Static Earth sensor, left, and conical scanning Earth sensor on the right.

Pitch is found by measuring the center of the chord with respect to an internal reference and roll is found from the length of the chord. If the scan angle is set properly the scan will traverse the northern or southern hemisphere of the Earth resulting in a one-to-one relationship between chordwidth and attitude. This will not be the case for large attitude errors.

```

1 EarthSensorScanning('initialize', [], 42167);
2 EarthSensorScanning('set_noise', 0, 42167);
3
4 clear d
5 d.rECI = [0;0;-42167];
6 a = linspace(-8,8,500);
7 x = zeros(2,500);
8 for k = 1:length(a)
9     d.qECIToBody = [CosD(a(k));SinD(a(k));0;0];
10    EarthSensorScanning('update', d);
11    t = EarthSensorScanning('get_output', d);
12    x(1,k) = t(1);
13    x(2,k) = t(2);
14 end
15 yL = {'Roll_(counts)','Pitch_(counts)'};
16 Plot2D( a, x, 'Roll_(deg)', yL, 'Earth_Sensor')
17
18 %% Scanning Earth sensor function
19 function t = EarthSensorScanning( action, d,
    rStandard )
20
21 persistent s p
22
23 switch action
24
25     case 'initialize'
26         if( nargin < 2 )
27             d = [];
28         end
29
30         if( isempty(d) )
31             p = Default;
32         else
33             p = d;
34         end
35         p.x = Unit( Cross( p.uScan, p.
            boresight ) );
36         s.scan = [1 1];
37         s.rollCounts = 0;
38         s.pitchCounts = 0;
39         s.valid = 0;
40         p.earthRadius = 6378 + 40; % CO2
41         p.cant = pi/2 - p.scanAngle;
42         s.failure = 0;
43         rho = asin( p.earthRadius /
            rStandard );
44         chord = Chordwidth( p.uScan, rho, p.
            cant, [0;0;-1] );
45         s.standardChord = 0.5*(chord(1)+chord(2));
46
47     case 'set_noise'
48         p.sigmalChord = d;
49
50     case 'update'
51         rho = asin( p.earthRadius / Mag( d.rECI )
            );
52         uNadir = -QForm( d.qECIToBody, Unit( d.rECI )
            );
53         noise = p.sigmalChord*randn(1,2);
54         chord = Chordwidth( p.uScan, rho, p.cant,
            uNadir ) + noise;
55
56         % Pitch measurement
57         pitch = atan2( uNadir'*p.x, uNadir'*p.
            boresight );
58
59         % Process the chordwidth data
60         s.valid = 1;
61         if( chord(1) > 0 && chord(2) > 0 && s.scan(1)
            && s.scan(2) )
62             s.standardChord = s.standardChord/2 + (
                chord(2) + chord(1))/4;
63             roll = (chord(2) - chord(1))/2;
64             pitch = pitch + 0.5*(noise(1) +
                noise(2));
65         elseif( chord(2) > 0 && s.scan(2) && s.
            standardChord > 0 )
66             roll = chord(2) - s.
                standardChord;
67             pitch = pitch + noise(2);
68         elseif( chord(1) > 0 && s.scan(1) && s.
            standardChord > 0 )
69             roll = s.standardChord - chord
                (1);
70             pitch = pitch + noise(1);
71         else
72             roll = 0;
73             s.valid = 0;
74         end
75
76         % Quantize
77         if( s.failure )
78             roll = 0;
79             pitch = 0;
80         end
81         s.rollCounts = round( roll/p.quantization )
            ;
82         s.pitchCounts = round( pitch/p.quantization )
            ;
83
84     case 'get_output'
85         t = [s.rollCounts; s.pitchCounts; round( s.
            standardChord/p.quantization ); s.
            valid];
86
87     end
88
89     end
90
91 %% Chordwidth calculation
92 function w = Chordwidth( u, rho, cant, nadir )
93
94     cNadir = u'*nadir;
95     sNadir = sqrt( (1-cNadir)*(1+cNadir) );
96
97     arg = (cos(rho)-cNadir*cos(cant))./(sNadir*sin
        (cant));
98
99     k = find(abs(arg) < 1 );
100    w = [0 0];
101    w(k) = acos(arg(k));
102
103 end
104
105 %% Chordwidth calculation
106 function p = Default
107
108     p.scanAngle = 5.72*(pi/180)*[-1 1];
109     p.uScan = [0;1;0];
110     p.quantization = 0.01*pi/180;
111     p.sigmalChord = 0.001*pi/180;
112     p.boresight = [0;0;1];
113
114 end

```



Example 11.2: Scanning Earth sensor with standard chord.

11.4.3 Scanning Earth sensor

One type of Earth sensor is a scanning sensor with two scanning elements. The Earth edges are measured directly in degrees relative to the sensor null. The sensor calculates the actual chordwidth that will be seen by the North and South scans, and the dihedral angle, or angular separation of the sensor null from the center of the Earth. The dihedral angle is essentially the same as the pitch angle. The geometry that is the basis for the calculations is shown in Fig. 11.3. The positions of the scans are defined by their cant angles or angular separation from the sensor spin axis.

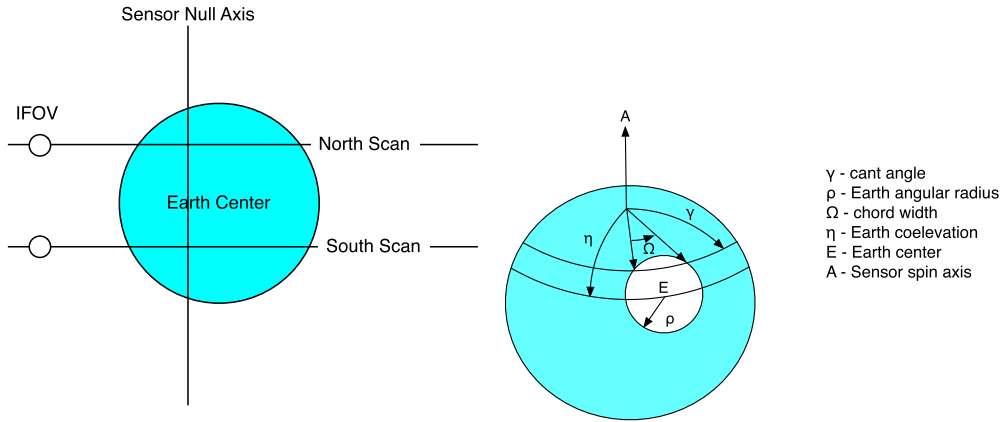


Figure 11.3 Diagram of the Earth-sensor scanning and chordwidth geometry.

$$\Omega = 2 \cos^{-1} \left(\frac{\cos \rho - \cos \eta \cos \gamma}{\sin \eta \sin \gamma} \right) \quad (11.8)$$

$$\cos \eta = \vec{A} \cdot \vec{E} \quad (11.9)$$

The sensor scanning uses back and forth scans to calculate each roll and pitch measurement. Fig. 11.4 shows the measured quantities.

Roll and pitch are calculated from these quantities according to:

$$\text{pitch} = \frac{1}{8} \sum_{j=1}^2 (-L_{j1} + L_{j2} + L_{j3} - L_{j4}) \quad (11.10)$$

$$\text{roll} = \frac{1}{4} \sum_{j=1}^4 (L_{2i} - L_{1i}) \quad (11.11)$$

The sensor calculates L_{ji} from the actual chordwidth and dihedral angles, and then computes roll and pitch in the same manner as the sensor, using integer math.

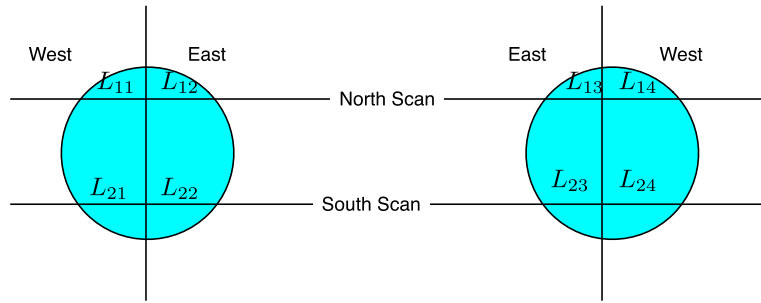


Figure 11.4 Nomenclature for measured chord parts.

The sensor also does some preprocessing of the roll and pitch angles. Depending on the number of edges sensed on each scan, roll, and pitch can be “latched” to the minimum or maximum output value or to zero to prevent erroneous output. A simple glitch filter also processes the measured values, which holds the output constant whenever there is a transition to or from the maximum or minimum value or a value latched to zero.

11.4.4 Analog Sun sensors

Analog Sun sensors are used in pairs to get attitude information. The sensor geometry is illustrated in Fig. 11.5.

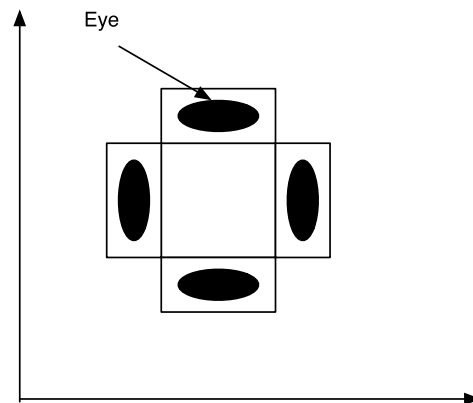


Figure 11.5 Sun-sensor geometry.

Two sensors' boresights are in the γz -array plane the other two are in the xz -array plane. The first is canted 45 deg from $+y$ and the other 45 deg from $+z$. The difference between the outputs of the sensors is an indication of the angle. Each sensor measures,

to first order, the cosine of the angle between the Sun vector and the sensing element. The measurement from one sensor is (in its simplest form)

$$\gamma = g \cos \theta \quad (11.12)$$

$$\cos \theta = n^T u_{Sun} \quad (11.13)$$

where g is a gain. Generally, analog sensors are modeled as

$$\gamma = \sum_{k=0}^N g_k \cos^k \theta \quad (11.14)$$

Since a single sensor is just a cosine detector it cannot give direction. If two such sensors are mounted together as shown in Fig. 11.6 the direction can be found.

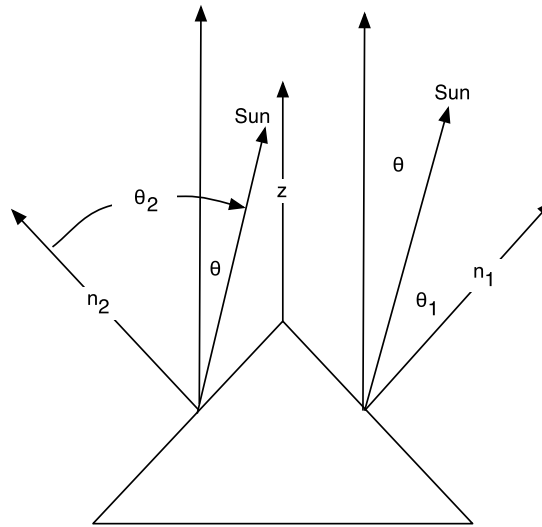


Figure 11.6 Simple one-dimensional Sun sensor.

$$\theta_1 = \cos^{-1}(n_1^T u_{Sun}) \quad (11.15)$$

$$\theta_2 = \cos^{-1}(n_2^T u_{Sun}) \quad (11.16)$$

If the angle between z and the sensor normal is β then

$$\theta_1 = \theta + \beta \quad (11.17)$$

$$\theta_2 = \beta - \theta \quad (11.18)$$

so that

$$\theta = \frac{\theta_2 - \theta_1}{2} \quad (11.19)$$

which gives sign information.

11.4.5 Digital Sun sensors

Digital Sun sensors measure the position of the Sun image in the sensor plane. These may use special patterns of thermal detectors in the sensor plane or may use CCD-type elements.

11.5. Gyros

Noise in a gyro or any physical system can be classified as:

1. White Gaussian;
2. Integrating white Gaussian produces random walk;
3. Integrating random walk produces random run;
4. By differentiating white noise we get phase noise;
5. Flicker noise is a Markov model

$$\dot{x} + \frac{1}{\tau}x = n \quad (11.20)$$

where τ is a time constant and n is Gaussian white noise;

6. By differentiating flicker noise we get flicker phase noise.

Allan Variance was initially created to study the frequency stability of oscillators. It is used for sensors to measure the time domain stability of a time signal due to noise. The Allan Deviation is the square root of the Allan Variance. Fig. 11.7 shows the Allan Deviation for the different types of noise.

The STIM300 is a MEMS gyro with noise parameters given in Table 11.2.

Table 11.2 STIM300 MEMS gyro noise.

Parameter	Value
Bias Run–Run	4 deg/h
Drift Rate Stability	3 deg/h
Bias Instability	0.3 deg/h
Angular Random Walk	0.15 deg/ $\sqrt{h}$

Bias Run–Run is the variation of the bias between runs. This, along with the bias, is removed by calibration when the gyro is turned on. Angular random walk is white noise on the rate. This causes the angle, which is the integral of rate, to have a random-walk

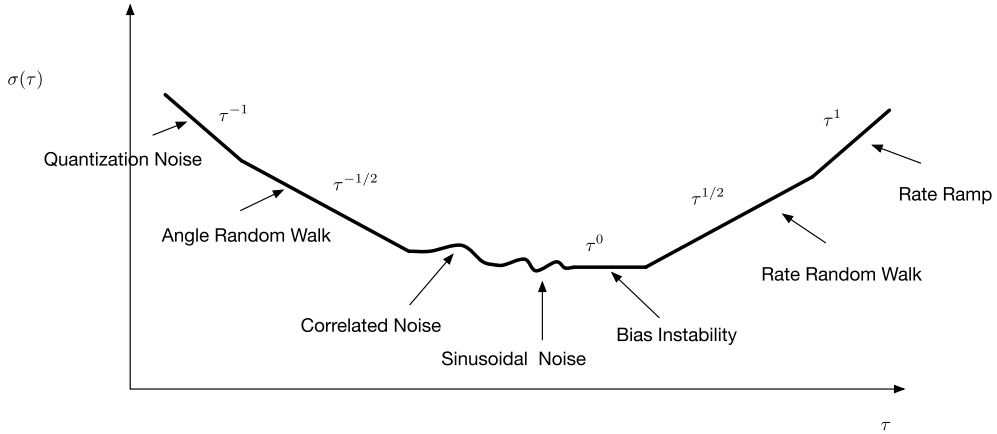


Figure 11.7 Allan Deviation and noise. τ is the averaging bin size.

element. The bias of a MEMS gyro wanders over time due to flicker in the electronics. Flicker noise is typically pink, that is it has a power spectral density that goes as the inverse of frequency. The power spectral density is the measure of a signal's power content versus frequency. The power spectral density of white noise is independent of frequency.

Table 11.3 summarizes the types of noise and how they match the MEMS gyro.

In Example 11.3 flicker pink noise is modeled by a first-order filter

$$y_k = y_{k-1} - \sigma y_{k-1} + \sigma u_k \quad (11.21)$$

where $\sigma = 0.1$ in this case.

The model that applies to the STIM300 single-degree-of-freedom gyro is

$$\dot{\theta} = \omega + b + f + \eta_\theta \quad (11.22)$$

$$\dot{f} = -\gamma f + \eta_f \quad (11.23)$$

$$\dot{b} = \eta_b \quad (11.24)$$

where θ is the integrated gyro angle, ω is the true inertial rate in body axes, b is the gyro bias, f is the bias instability, η_θ causes the integrated attitude to random walk, and η_b causes the bias to drift. The output of the gyro is both angular rate and, if it is a rate-integrating gyro, angular increments. The noise number for random walk, η_r is given in $\text{deg}/\sqrt{\text{h}}$. The conversion to a 1-sigma noise number is

$$n = \left(\frac{\pi}{180} \right) \left(\frac{\eta_r}{60} \right) \quad (11.25)$$

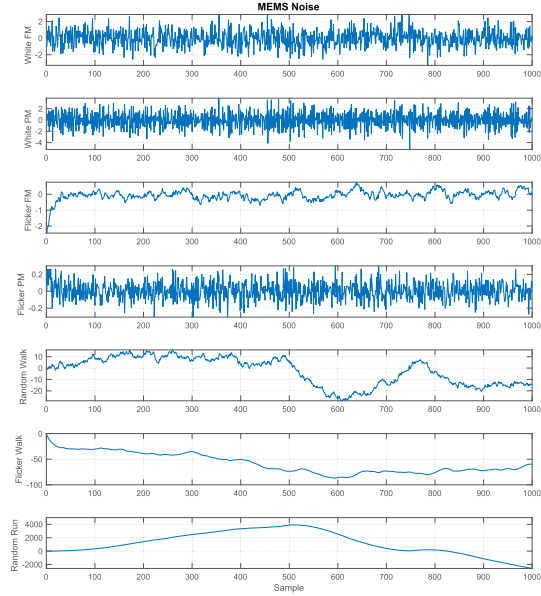
Table 11.3 Gyro noise with frequency and time bin size dependence. “P” means phase and “F” frequency.

Noise	Spectrum	Allan Deviation	Alan Variance	Modified Allan Deviation	Modified Allan Variation	Time Deviation	Time Variation	MEMS
White PM	f^2	τ^{-1}	τ^{-2}	$\tau^{-3/2}$	τ^{-3}	$\tau^{1/2}$	τ^{-1}	
Flicker PM	f	τ^{-1}	τ^{-2}	τ^{-1}	τ^{-2}	τ^0	τ^0	
White FM	f^0	$\tau^{-1/2}$	τ^{-1}	$\tau^{-1/2}$	τ^{-1}	$\tau^{1/2}$	τ^1	
Flicker FM	f^{-1}	τ^0	τ^0	τ^0	τ^0	τ^1	τ^2	Bias Instability
Random Walk	f^{-2}	$\tau^{1/2}$	τ^1	$\tau^{1/2}$	τ^1	$\tau^{3/2}$	τ^3	Angular Random Walk
Flicker Walk	f^{-3}	τ^1	τ^2	τ^1	τ^2			Drift-Rate Stability
Random Run	f^{-4}	$\tau^{3/2}$	τ^3	$\tau^{3/2}$	τ^3			

```

1 %% MEMS noise
2
3 yL = { 'White_FM' 'White_PM' 'Flicker_FM' ...
4         'Flicker_PM' 'Random_Walk' ...
5         'Flicker_Run' 'Random_Run' };
6
7 n = 1000;
8 nS = zeros(7,n);
9
10 sF = 0.1;
11
12 nS(1,:) = randn(1,n);
13 nS(2,:) = [0, diff(nS(1,:))];
14 nS(3,:) = Filter(sF, randn(1,n));
15 nS(4,:) = [0, diff(nS(3,:))];
16 nS(5,:) = cumsum(nS(1,:));
17 nS(6,:) = cumsum(nS(3,:));
18 nS(7,:) = cumsum(nS(5,:));
19
20 Plot2D(1:n, nS, 'Sample', yL, 'MEMS_Noise')
21
22 % A filter to simulate pink noise
23 function y = Filter(sig,u)
24
25 n = length(u);
26 y = zeros(1,n);
27 y(1) = u(1);
28
29 for k = 2:n
30     y(k) = y(k-1) - sig*y(k-1) + sig*u(k);
31 end
32
33 end

```



Example 11.3: Noise sources.

Angular increments can be converted to rates using one of the following formulas [1],

$$\omega_k = \frac{a_k}{\Delta T} \quad (11.26)$$

$$\omega_k = \frac{3a_k - a_{k-1}}{\Delta T} \quad (11.27)$$

$$\omega_k = \frac{11a_k - 7a_{k-1} + 2a_{k-2}}{\Delta T} \quad (11.28)$$

where a_k is the angular increment at step k , that is

$$a_k = \theta_k - \theta_{k-1} \quad (11.29)$$

and ΔT is the time step or the time interval between step k and step $k - 1$.

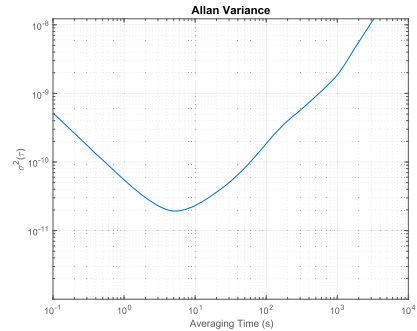
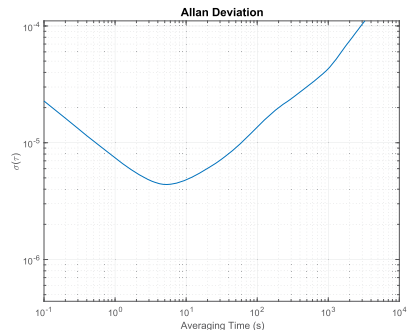
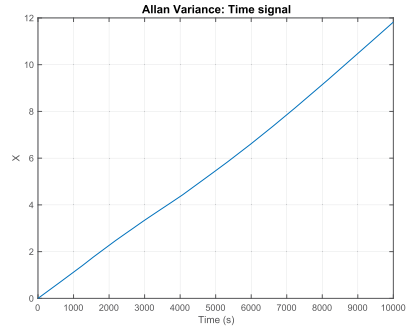
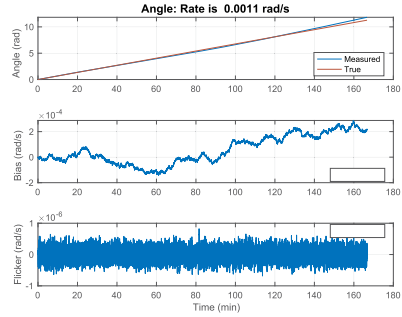
Example 11.4 shows simulation results, Allan Variance, and Allan Deviation for the STIM300 model. The drift-rate stability, bias instability, and angular random walk are clearly shown in the plots.

Note that the MEMS gyro has a large bias. This must be measured when the gyro is turned on and stored as a constant. This should be done before the release of the spacecraft from the ISS.

```

1 %% Simulate a MEMS gyro
2 el = ISSOrbit;
3 degPHrToRadPs = pi/180/3600;
4 n = 1e5; % Number of steps
5 dT = 0.1; % sec
6 d = RHSMEMSGyro;
7 d.rate = 2*pi/Period(el(1)); % rad/s
8 d.angleNoise = RW2SDev(0.15);
9 d.flickerNoise = 0.3*degPHrToRadPs;
10 d.biasNoise = 3*degPHrToRadPs;
11
12
13 %% Run the simulation
14 xP = zeros(3,n);
15 x = zeros(3,1);
16
17 for k = 1:n
18     xP(:,k) = x;
19     x = RK4(@RHSMEMSGyro,x,dT,0,d);
20 end
21
22 t = (0:n-1)*dT;
23 [tS,tL] = TimeLabl(t);
24
25 yL = {'Angle_(rad)', 'Bias_(rad/s)' 'Flicker_(rad/s)'};
26
27 s = sprintf('Angle: %f rad/s', d.rate);
28
29 IL = {'Measured', 'True'};
30
31 Plot2D(tS,[xP;d.rate*t],tL,yL,s,'lin',{'[1_4]','[2]','[3]'},[1,1,1],[1,1,1],[IL,{'',''}]);
32
33 %% Allan variance
34 AllanVariance(xP(1,:),t);
35
36 Figui

```



Example 11.4: STIM300.

11.6. Other sensors

11.6.1 Magnetometers

Magnetometers are used for attitude determination and for measuring the magnetic field for adjusting the gain of magnetic-torquer systems. Magnetometers measure the total magnetic field and must be kept away from sources of magnetic fields. Usually, they are mounted on booms.

The magnetic-torque rods will produce magnetic fields that will be seen by the magnetometer. Example 11.5 shows the effect on the measured magnetic field. Two cases are shown. One has the magnetometer at the opposite corner of the bus. The second puts it on a 25-cm boom along the $-Y$ direction. Putting the magnetometer on the boom reduces the torquer field by half. Given that the field is deterministic, we can measure it during integration and test and subtract it whenever a torquer is turned on. The magnetic field due to a dipole is

$$b = \frac{\mu_0}{4\pi} \left(\frac{3r(M^T r)}{|r|^5} - \frac{M}{|r|^3} \right) \quad (11.30)$$

This formula assumes an infinitesimal dipole so it is only applicable when the point is a distance from the current loop. The field drops off as $|r|^3$ so its effect can be mitigated by distance. For this reason, magnetometers are often placed on booms. Well-known formulas exist for the magnetic field for a finite solenoid, [2], that would apply to a torquer bar.

Example 11.5 shows the effect of a magnetic dipole on the measured magnetic field in an ISS orbit. The first plot shows the field around a dipole. The second shows the measured magnetic field. The field due to the torquer can be measured and subtracted from the measurement. A bandpass filter, with the center frequency at orbit rate, can also be used to reduce errors.

There will be other sources of magnetic fields, such as current loops due to the spacecraft power bus and the motors in the reaction wheels.

11.6.2 Accelerometers

Accelerometers measure the acceleration of a mass relative to their base. A simple proof-mass accelerometer is shown in Fig. 11.8.

The mass is attached to the base via a spring and damper. The transfer function is

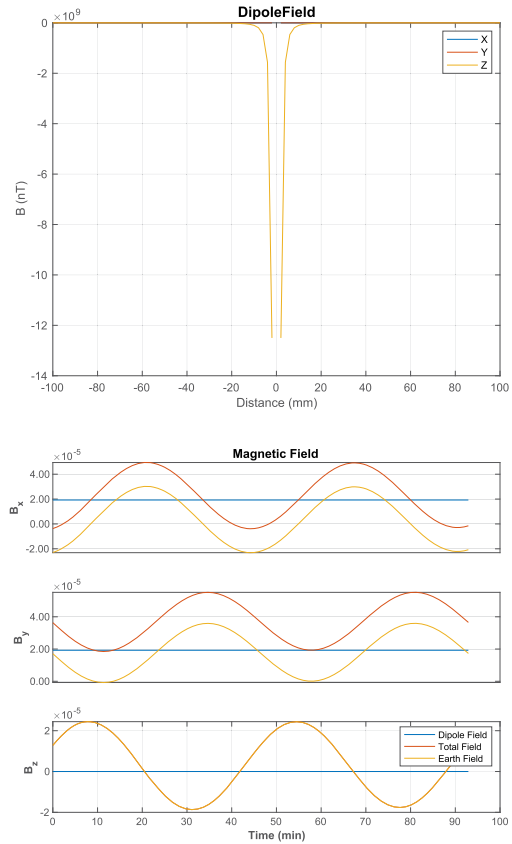
$$\frac{s^2}{s^2 + 2\zeta\omega s + \omega^2} \quad (11.31)$$

At low frequencies, the accelerometer has no response. At high frequencies, it has a transfer function of unity.

```

1 %% Local magnetic field
2
3 r = [linspace(-0.1,0.1,101);zeros(2,101)];
4
5 b = MagneticFieldDipole([0;0;1],r);
6
7 Plot2D(r(1,:)*1000,b*1e9,'Distance(mm)','Bz(nT)')
8 , 'DipoleField')
9
10 % All three torquers are on
11 dipole = 1; % AIM^2
12
13 % Unit vectors for the dipoles
14 u = eye(3);
15
16 % Points on corner of spacecraft and on a boom
17 p = [10;10;10]*0.01;
18
19 % ISS Orbit
20 [el,jD0] = ISSOrbit;
21 [r,~,t] = RVOrbGen(el);
22
23 % Magnetic field
24 jD = jD0 + t/86400;
25 bEarth = BDipole(r,jD);
26 b = MagneticFieldDipole([0;0;1],p);
27
28 bT = bEarth + b;
29 yL = {'B_x' 'B_y' 'B_z'};
30 n = length(jD);
31
32 TimeHistory(t,[b*ones(1,n);bT;bEarth],yL,'
33             MagneticField',{[1 4 7] [2 5 8] [3 6 9]})
34 ;
35 legend('DipoleField','TotalField','Earth
36 Field')

```



Example 11.5: Effect of the magnetic-torque rods on the measured magnetic field.

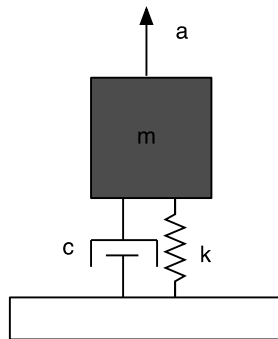


Figure 11.8 Proof-mass accelerometer.

11.6.3 Potentiometers

Potentiometers measure angles by changes in resistance. They usually are low-accuracy sensors and may have a large deadband around zero. Many produce a signal when the device passes through zero, which is more accurate than the angle measurements. Usually, potentiometers are used as a backup for other measurements, such as counting steps from a stepping motor.

11.6.4 Angle encoders

Angle encoders are either absolute or relative. Absolute angle encoders give you the actual measurement from a fixed reference. Relative give you the change in angle. Optical angle encoders are often made from a pattern painted on a glass disk. Encoders are expensive but are necessary for some high-precision applications, such as sinusoidal motor control.

11.7. Star cameras

11.7.1 Pinhole camera

The lowest-order approximation to the optical system is a pinhole camera. The pinhole camera is shown in Fig. 11.9.

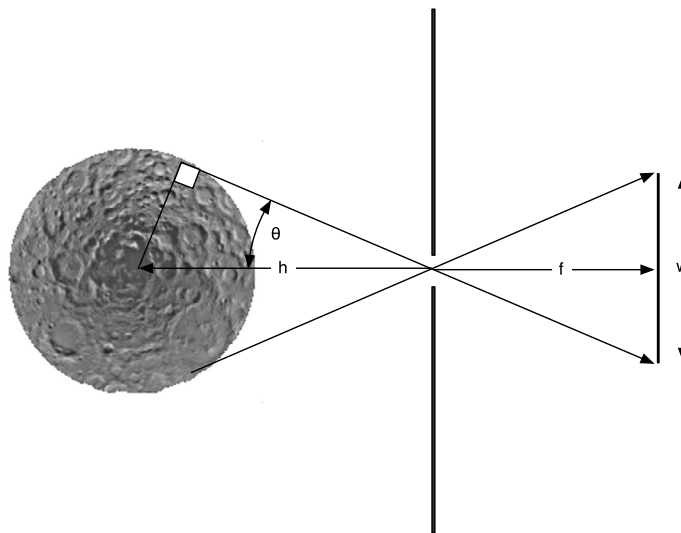


Figure 11.9 Pinhole camera.

A point $P(X, Y, Z)$ is mapped to the imaging plane by the relationships

$$u = \frac{fX}{Z} \quad (11.32)$$

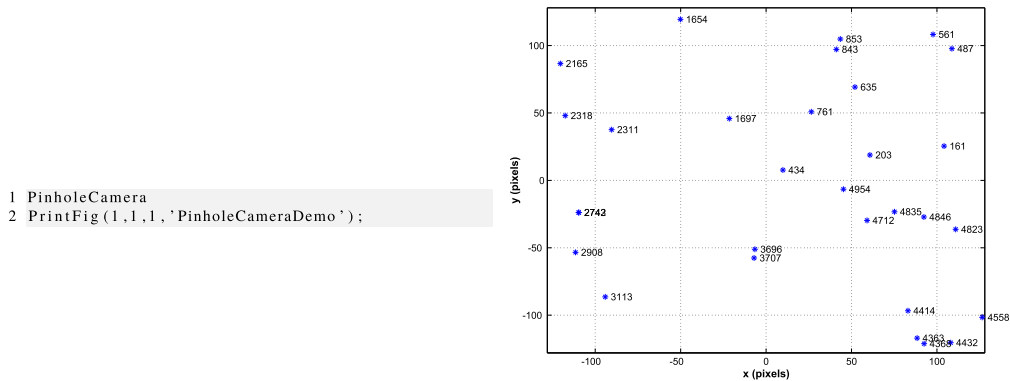
$$v = \frac{fY}{Z} \quad (11.33)$$

where u and v are coordinates in the focal plane, f is the focal length, and Z is the distance from the pinhole to the point along the axis normal to the focal plane. This assumes that the Z -axis of the coordinate frame X, Y, Z is aligned with the boresight of the camera.

The angle that is seen by the imaging chip is

$$\theta = \tan^{-1} \left(\frac{w}{2f} \right) \quad (11.34)$$

where f is the focal length. The shorter the focal length, the larger the image. The pinhole camera does not have any depth of field, but that is unimportant for many far-field imaging problems. The field-of-view of a pinhole camera is limited only by the sensing element. An example of a pinhole image from the Hipparcos catalog (showing only stars with a visual magnitude < 6) is given in Example 11.6.



Example 11.6: Pinhole-camera star image.

Errors associated with the star camera are shown in Fig. 11.10. These errors are discussed in the following sections.

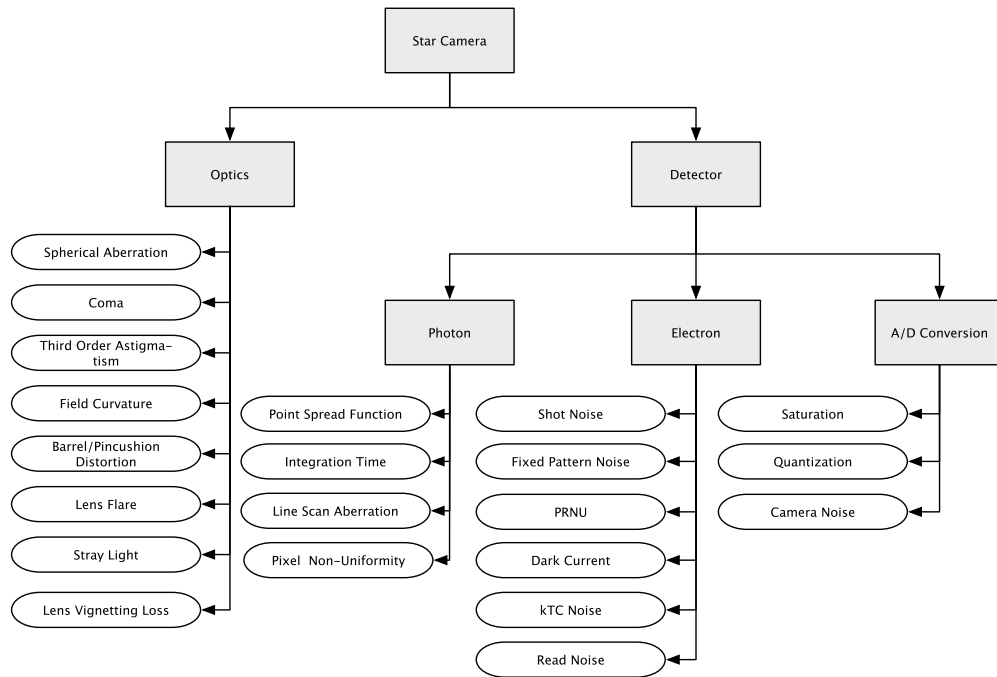


Figure 11.10 Star-camera errors.

11.7.2 Optical errors

Optical errors in a camera are due to the telescope barrel and lens. Optical errors are given in Table 11.4.

Table 11.4 Optical-error budget terms.

Error Source	Bias	Long	Orbit Rate	Short	Description
Spherical aberration	×			×	This is the difference in focal length between axial rays and rays along the periphery of aperture.
Coma	×			×	This is a geometric property in which offaxis point sources do not all come together at the same point.
Third-order astigmatism	×			×	This is a geometric effect even in symmetric optical systems.
Field curvature	×			×	The image really lies on the surface of a sphere, not a flat plate. This causes offaxis errors when an imaging chip is used.

continued on next page

Table 11.4 (continued)

Error Source	Bias	Long	Orbit Rate	Short	Description
Barrel/Pin-cushion distortion	×			×	Magnification decreases with distance from the optical axis.
Lens flare				×	Light scattered in a telescope by internal reflection in lens.
Stray light				×	Scattering when a bright source, such as the Sun, is in the field of view. Diffraction features are caused by support structures, baffles, and other objects.

11.7.3 Imaging-chip errors

Imaging-chip errors are given in Table 11.5 [3]. These errors can be mitigated by

1. Varying the pixel integration time;
2. Combining frames;
3. Calibrating.

Table 11.5 Imaging-chip error budget terms.

Error Source	Bias	Long	Cyclic	Short	Description
Point Spread Function				×	Errors in the point spread function will reduce the fitting accuracy.
Integration Time				×	Long integration times can cause overexposure resulting in saturated pixels.
Line-Scan Aberration				×	A time delay from when the top and bottom rows are integrated due to rolling shutters.
Pixel Nonuniformity	×				This will cause a bias since the actual light measured is only over a portion of the pixel.
Shot Noise				×	Shot noise is caused by the random arrival of photons. This is a fundamental trait of light. Since each photon is an independent event, the arrival of any given photon cannot be precisely predicted; instead the probability of its arrival in a given time period is governed by a Poisson distribution.
Fixed-Pattern Noise				×	The is temporally constant pixel nonuniformity in noise.

continued on next page

Table 11.5 (continued)

Error Source	Bias	Long	Cyclic	Short	Description
PRNU				×	Photo Response Nonuniformity cause errors since each pixel does not produce the same response for the same number of photons.
Dark Current				×	This is current that flows in the pixel in the absence of light. It increases with time.
kTC Noise				×	Thermal noise.
Read Noise				×	These are onchip sources of noise that are combined into the Gaussian noise generated when a pixel is read.
Saturation				×	If a pixel saturates it will cause an error in the centroiding fit. This is a problem if multiple pixels in a centroid are saturated.
Quantization				×	Limits the accuracy of the centroiding computation due to minimum bit value.
Camera noise				×	All sources of noise that come between the readout of the pixel and the A/D conversion.

PRNU and fixed-pattern noise are characterized as random noise but they remain constant for each image once the detector is turned on.

11.8. GPS

GPS receivers can use any one of several navigation-satellite constellations. Four are

1. Global Positioning Systems (US);
2. GLONASS (Russia);
3. BeiDou (PRC);
4. Galileo (ESA).

Many receivers use all of them. Most GPS receivers provide a navigation solution but often allow access to the raw data. For attitude determination, raw signals are required. One or more antennas are required. When multiple antennas are used, algorithms use GPS differential carrier-phase measurements between the antennas to determine the attitude [4]. Antenna geometry is an important factor in accuracy.

References

- [1] E. Wong, W. Breckenridge, Inertial attitude determination for dual-spin planetary spacecraft, *Journal of Guidance* 6 (6) (1983) 491–498.
- [2] E.E. Callaghan, S.H. Maslen, The magnetic field of a finite solenoid, <https://www.osti.gov/biblio/4121210>, 10 1960.

- [3] M. Knutson, D. Miller, Fast Star Tracker Centroid Algorithm for High Performance CubeSat with Air Bearing Validation, Master's thesis, M.I.T. Department of Aeronautics and Astronautics, June 2013, SSL # 5-12.
- [4] Charles Wang, Rodney A. Walker, Werner Enderle, Single antenna attitude determination for FedSat, in: Proceedings Institute of Navigation GPS-02, 2002, pp. 1-12.